

SMARTROUTEADMM : AI DRIVEN ROUTE SCHEDULING AND ROUTE OPTIMIZATION USING ADMM

PROJECT REPORT

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**BACHELOR OF TECHNOLOGY
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**COMPUTER SCIENCE ENGINEERING - ARTIFICIAL INTELLIGENCE
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**COMPUTER SCIENCE ENGINEERING - ARTIFICIAL
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BONAFIDE CERTIFICATE

This is to certify that the report entitled "**SMARTROUTEADMM : AI driven Route scheduling and Route optimization using ADMM**" submitted by **Anto Rishath** (Register Number- **CB.SC.U4AIE23103**), **Arunoday A K** (Register Number- **CB.SC.U4AIE23106**), **Achutharaj R** (Register Number- **CB.SC.U4AIE23108**), **Vijay Santhosh P** (Register Number- **CB.SC.U4AIE23157**), **Yashwanth P T** (Register Number- **CB.SC.U4AIE23164**) in partial fulfillment of the requirements for the courses **22AIE211- Introduction to Communication & IoT** and **22AIE213 Machine Learning** for the award of the **Degree of Bachelor of Technology** in the "**COMPUTER SCIENCE ENGINEERING - ARTIFICIAL INTELLIGENCE**" is a bonafide record of the work carried out by them under our guidance and supervision at Amrita School of Artificial Intelligence, Coimbatore.

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*Submitted for the university examination held on **16.04.2025***

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DECLARATION

We, **Anto Rishath** (Register Number- CB.SC.U4AIE23103), **Arunoday A K** (Register Number- CB.SC.U4AIE23106), **Achutharaj R** (Register Number- CB.SC.U4AIE23108), **Vijay Santhosh P** (Register Number- CB.SC.U4AIE23157), **Yashwanth P T** (Register Number- CB.SC.U4AIE23164) hereby declare that this thesis entitled ”**SMARTROUTEADMM : AI driven Route scheduling and Route optimization using ADMM**”, is the record of the original work done by us under the guidance of **Dr. Sundaresan S & Dr. Anand Kumar**, Amrita School of Artificial Intelligence, Coimbatore. To the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/ associateship/fellowship/or a similar award to any candidate in any university.

Place:

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List of Abbreviations

AI	Artificial Intelligence
ADMM	Alternating Direction Method of Multipliers
LSTM	Long Short Term Memory
IR	Infrared

Abstract

Efficient allocation of buses and scheduling of drivers and conductors is a persistent challenge in modern urban transportation. Traditional systems often rely on static, manual scheduling, resulting in inefficient resource use, longer passenger waiting times, and inconsistent service, especially during peak hours or fluctuating demand.

To address this, we introduce SmartRoute ADMM, an AI-driven framework for dynamic bus rerouting and rescheduling. Using artificial intelligence (AI) and the Alternative Direction Method of Multipliers (ADMM), SmartRoute ADMM adapts to real-time passenger demand, optimizing bus routes, arrival times, and backup deployments. The system employs a three-dimensional space-time network, where AI predicts future passenger demand, enabling proactive route planning.

By decomposing the optimization problem with ADMM, the system achieves scalability and efficiency in large transit networks. We evaluated the method on an n -node bus network, demonstrating significant improvements in service quality, resource utilization, and passenger satisfaction. SmartRoute ADMM offers an intelligent, adaptable solution for urban transit systems facing unpredictable demand.

Chapter 1

Introduction

Transportation is one of the most important sector which helps in the development of the economy. When compared to the previous the rate of transportation is rapidly increasing through out the years whether it is a public mode of transportation or private mode of transportation. There are lot of people who are still preferring public modes of transportation especially for buses especially in urban areas. Because of which results in traffic congestion has become a serious phenomenon and increasing attention is paid to environmental traffic. Buses play a crucial role in this transportation sector.

For example, the people those who are working are still preferring buses and many students and different people who wants to go for some immediate place in time. Many studies such as wang et al, Babagi and saydi Patamaadi, Kim et al and Tong et al scheduled buses considering different situations including traffic congestion, single destination, multiple destination and customized bus service design to ensure well designed bus routes and time tables and to improve service quality. To ensure the safety and security which complies with traffic rules, there is a number of passengers that a bus can accommodate which is called a bus capacity. These bus capacities would always

fluctuate like it may increase or decrease. For example: During holidays time there would be a increase in passengers in different bus stops when compared with a normal day, because of which it leads to congestion and the buses cannot withstand the capacity of the passengers which leads to a poor service quality. In this context bus rerouting & rescheduling can be implemented to reduce the passenger density.

In practice, public transportation systems are unstable and changeable due to uncertain passenger flow. During the planning process the operators will be manually scheduling the bus routes and the bus timetables for the drivers and conductors based on the average and estimated passenger flow to guarantee service. But in real time scenario the passenger flow may increase or decrease due to which they force the operators should reroute and reschedule the buses which utilizes lot of resources and provides a poor service for the passenger and elegant for the operator to change the routes and schedule everytime manually. In this case if the bus follows the original routes there must be some passengers that cannot board and some buses that have vacant seats leads to lower service quality and waste of capacity, respectively. Therefore an automated system must be present to adjust the original plan and generating rerouting and rescheduling in a more resource optimized manner which are important for addressing the passenger demand fluctuation conditions when the buses capacities are limited.

1.1 Literature Survey

Efficient bus routing and scheduling have been extensively studied to address urban transportation challenges such as traffic congestion, fluctuating passenger demand, and operational cost optimization.

Table 1.1: Summary of Literature on Bus Scheduling and Route Optimization

Study/Authors	Method Used	Focus Area	Key Outcomes
Prof. Priyanka Salunkhe et al.	Automated Scheduling System	Real-time scheduling, passenger-centric services	Improved route allocation, reduced delays, better timetable adherence
Cao and Ceder	Genetic Algorithm (GA)	Real-time rescheduling	Minimized total travel time and reduced number of active buses
Liu et al.	Genetic Algorithm (GA)	Traffic disruption scenarios	Reduced waiting time, in-vehicle time, and operational delays
Wang et al.	Genetic Algorithm (GA)	Congestion scenarios	Reduced operational costs under heavy traffic conditions
Li et al.	Auction-Based Algorithm	Dynamic schedule adjustment during disruptions	Minimized service delays while ensuring fair resource allocation
Nikolić and Teodorović	Bee Colony Optimization	Demand fluctuation management	Balanced supply-demand and reduced transportation costs

1.2 Problem Statement

Urban bus systems in densely populated areas face significant challenges in adapting to fluctuating passenger demand across different time periods, such as peak versus off-peak hours or weekdays versus holidays. Traditional static and pre-planned scheduling approaches fail to address real-time variations, leading to issues like overcrowded buses,

underutilized capacity, long waiting times, and inefficient routing. These inefficiencies result in reduced service quality, passenger dissatisfaction, and increased operational costs. There is a critical need for a dynamic, automated, and demand-responsive bus management framework that can optimize routing, scheduling, and capacity utilization in real-time to enhance operational efficiency and passenger satisfaction.

1.3 Objectives

- **Dynamic Bus Scheduling and Routing:** Develop a demand-responsive bus management system that adapts to real-time or predicted passenger demand, ensuring efficient routing and scheduling.
- **Capacity Optimization:** Maximize the utilization of available bus capacity to address issues like overcrowding and underutilized buses.
- **Operational Cost Reduction:** Minimize operational costs by optimizing resource allocation and reducing delays.
- **Passenger Satisfaction:** Improve service quality by reducing waiting times, enhancing reliability, and ensuring a better travel experience for commuters.

1.4 Organization of the Report

The structure of this report is outlined below:

- **Chapter 1: Introduction**

Provides the background of urban transportation challenges, a review of existing

work in bus scheduling and demand prediction, the motivation behind the project, the problem statement, and the key objectives.

- **Chapter 2: Background**

Discusses the key technologies used including LSTM for time-series forecasting, ADMM optimization for rerouting and rescheduling, IR sensor-based passenger detection using Arduino Mega, and the ReactJS frontend for real-time data visualization.

- **Chapter 3: Proposed Work**

Describes the design of the physical network and its transformation into a space-time-state model, development of the demand-aware scheduling system using ADMM, and the complete integration of hardware and software components.

- **Chapter 4: Results and Discussion**

Presents the outcomes of LSTM-based forecasting, ADMM optimization outputs, results from the hardware implementation for passenger detection, and frontend dashboard screenshots. Graphs and visual outputs are discussed to demonstrate system performance.

- **Chapter 5: Conclusion and Future Work**

Summarizes the contributions of the project, evaluates its effectiveness in real-world-like scenarios, and outlines potential improvements and future directions for deployment and scalability.

Chapter 2

Background

2.1 Introduction

Transportation is a key sector fueling economic growth, and public transport, particularly buses, is essential in urban centers. With growing usage, particularly during peak hours like holidays, variable passenger demand typically results in congestion and suboptimal quality of service owing to insufficient bus capacity. Manual scheduling using traditional average passenger flow-based methods is unable to respond to real-time changes, leading to underutilization or overloading of buses. This inefficiency brings to the forefront the necessity for automated systems capable of dynamically redirecting and rescheduling buses as a reaction to demand changes, maximizing resources, and enhancing overall service quality.

2.2 Data Acquisition

Since actual datasets of passenger flows at bus stops or routes are not available, the data was synthetically created in order to represent typical passenger demand behavior in different time periods. The data creation process was designed to replicate reality for features like rush hour, weekday-weekend contrast, and random increases in demand (e.g., holidays or sporting events). This meant generating time-series data for every bus stop within a specified timeframe (e.g., daily counts for a number of months), with the passenger number being driven by temporal variables such as the day of the week, hour of the day, and special days.

To simulate realism in the artificial dataset, Gaussian noise was included to model random fluctuations in passenger counts, and periodic functions were employed to simulate daily commuting patterns (e.g., morning and evening rush hours). This dataset served as the initial training material for predictive models that could learn temporal patterns in passenger flow.

2.3 Passenger Demand Forecasting

To enable dynamic bus rescheduling and rerouting, an LSTM neural network was employed to forecast the future passenger flow. LSTM is a recurrent neural network (RNN) that is best suited for time-series forecasting owing to its capacity to maintain long-term dependencies.

It was trained using artificial time-series data based on attributes such as previous

passenger history, hour, and day of the week. The architecture employed stacked LSTM followed by dense to predict future numbers of passengers by inputting past window values.

Performance was measured based on Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The LSTM model, after training, successfully predicted short-term passenger demand, which could serve as important input for automatic bus rerouting and rescheduling under changing demand, thus enhancing capacity utilization and service reliability.

2.4 3-Dimensional Space-Time-State Network

A key component of the proposed methodology is the construction of a 3-dimensional space-time-state network, which forms the mathematical foundation of the optimization model. This network integrates:

- **Space:** Represents the physical layout—bus stops, terminals, and route segments.
- **Time:** Captures operational dynamics such as departure/arrival times, service frequency, and passenger time windows.
- **State:** Encodes the status of buses and passenger allocation over time and space.

Bus movement is modeled through arcs—pickup, inbound, outbound, and drop-off—constrained by passenger time windows to ensure timely service. The model assigns passengers to buses based on destinations and network rules, optimizing routes

and schedules to minimize operational costs like travel time, distance, and service disruptions.

This network enables a comprehensive representation of both spatial and temporal dynamics, supporting efficient route planning under fluctuating demand.

2.5 ADMM Based Optimization

Optimization of the large-scale space-time-state network is computationally intensive because it is large in size and also because the problem is non-convex. To overcome this, the Alternating Direction Method of Multipliers (ADMM) is utilized. ADMM breaks down the entire problem into smaller, solvable sub-problems that are independent and parallelizable, dramatically enhancing computational efficiency.

Each of the sub-problems is a localized part of the transportation network—a route segment or a set of stops—and is solved iteratively. ADMM coordinates solving the solutions of the sub-problems to be globally consistent and system-wide optimal.

One of the key strengths of ADMM is its capacity to handle non-convex functions through conversion of these into convex form. This is made possible through linearization of complicated terms and added auxiliary variables, enabling each sub-problem to be convex and computationally convenient.

By application of decomposition and parallel computation, ADMM improves scalability, speeds up convergence, and facilitates real-time optimization of bus routing and scheduling and, thus, is ideally suited to large-scale, dynamic public transportation networks.

Chapter 3

Proposed Work

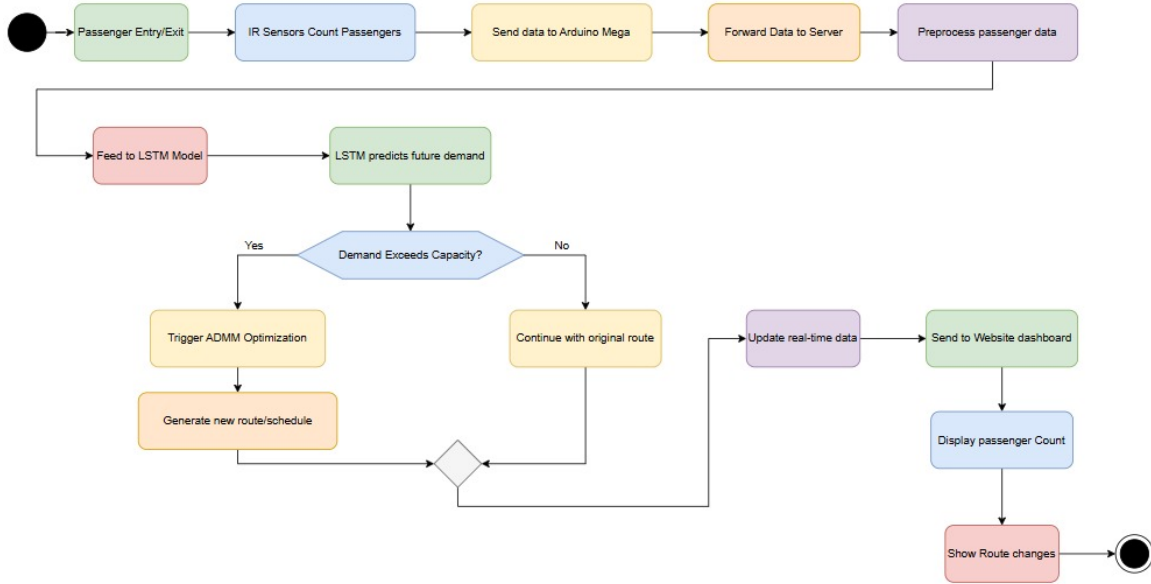


Figure 3.1: Complete Project Workflow

3.1 Dynamic Bus Routing Using IR Sensor Networks and ADMM Optimization

This project proposes a smart and adaptive bus scheduling and routing system which can dynamically adjust routes and re-shuffle the working hours of bus drivers and conductors to which bus they are allocated in response to the real-time count of the

passengers, which we consider as our primary constraint in this working hardware model for demonstration purposes. The system integrates infrared (IR) sensor-based passenger monitoring, which is designed in a graph structure with a decentralized optimization technique known as the Alternating Direction Method of Multipliers (ADMM), which could handle the large-scale problem and divide it into a small set of subproblems. This unique approach ensures operational efficiency, provides higher service quality for the passengers and responsiveness to fluctuating urban transit demands.

3.2 System Overview

The proposed system for demonstration purpose of our idea is composed of four main components:

- **IR Sensor Network** for real-time passenger count so that we could identify fluctuation of passenger demand at various bus stops.
- **Model Prediction** using LSTM model for predicting the number of passengers of the next day .so that the scheduling can be done based on passenger count.
- **ADMM Optimization** for the computation of decentralized and to find the most optimized routes.
- **Dynamic Bus Dispatching and Communication System** for real-time implementation where all the logic of the code is running in the backend and an user friendly display to view the schedules.

3.3 System Architecture

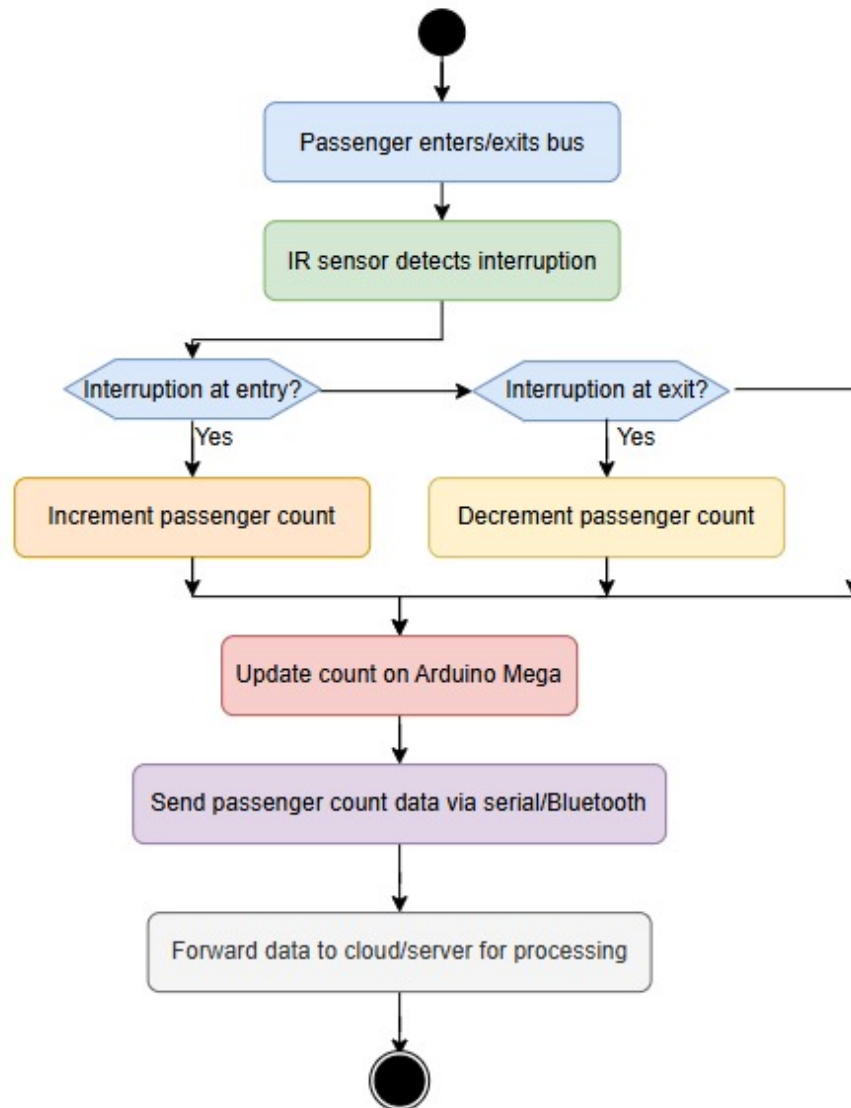


Figure 3.2: Hardware Workflow

3.3.1 IR Sensor Network as Virtual Bus Stops

To demonstrate a flexible bus stop infrastructure, six IR sensors are strategically deployed at each bus stop where the passage of the buses takes place, and in total, we are having 8 nodes such that 2 nodes represent the departure and the destination of the buses. These sensors function as *virtual bus stops* by detecting the number of passengers that are waiting in the bus busstops and also using this, we can detect the passenger demand fluctuations.

Each sensor captures passenger demand and transmits data wirelessly to an Arduino Mega, where the board transfers the data to the backend code where all the server logic takes place. The server aggregates sensor readings to assess spatial and temporal demand fluctuations.

3.3.2 Real-Time Passenger Monitoring

To keep an eye on passenger activity at every virtual stop, the IR sensor network runs constantly. These sensors allow the system to precisely differentiate between boarding and alighting events by counting the number of people entering and leaving particular areas. Understanding passenger flow and real-time demand management depend on this data.

The following are the tasks carried out by the monitoring system:

- **Activity for Boarding and Alighting:** Monitors passenger movement to distinguish between those boarding and disembarking from the bus.

- **Crowd Density Estimation:** Determines how many people are at each virtual stop over time in order to estimate the crowd density.
- **Finding Demand Patterns:** Allows for the identification of peak and off-peak times by identifying daily and hourly trends in passenger behavior.

3.4 Dynamic Routing Logic

3.4.1 Threshold-Based Route Adjustment

The system evaluates live passenger counts against set criteria. The routing logic gives priority to a particular stop when demand there surpasses the threshold and reroutes buses accordingly. It lessens passenger wait time, idle runs, and crowding.

3.4.2 ADMM-Based Optimization Algorithm

The routing problem is decomposed into multiple sub-problems:

- **Vehicle Sub-Problem:** Determines the optimal route and schedule for each bus.
- **Stop Sub-Problem:** Adjusts demand estimates and resource allocations for each stop.

These are solved in parallel and coordinated using ADMM, which ensures that each local solution respects global constraints while enabling fast convergence.

3.5 ADMM-Based Optimization Framework

3.5.1 Space-Time-State Network Model

By combining spatial, temporal, and operational state data, the Space-Time-State (STS) network model offers a thorough 3D framework to depict bus operations.

- **Space:** Represents the physical layout of the transport system, including stops, depots, and routes.
- **Time:** Discretizes the scheduling horizon, allowing tracking of bus movements and enforcing time window limits.
- **State:** Captures operational states such as boarding, alighting, idle times, and passenger load levels.

Every node in the STS network is defined by a tuple (i, t, w) , where:

- i is the spatial site,
- t is the time point, and
- w is the present passenger condition.

Transitions between nodes model changes in time and passenger load. This structure allows the system to optimize bus routes and schedules dynamically while respecting capacity, timing, and demand constraints.

3.5.2 Objective Function

The goal is to minimize total system cost:

$$\text{Minimize } \sum (C_{\text{transport}} + C_{\text{waiting}} + C_{\text{fixed}})$$

- $C_{\text{transport}}$: Fuel and operational costs.
- C_{waiting} : Total passenger waiting time.
- C_{fixed} : Labor, maintenance, and static costs.

3.5.3 Constraints

The optimization is subject to:

- Vehicle capacity limits such that the number of passengers that the bus can accommodate.
- Time windows for passenger pickups where we have to consider the situations where it takes extra time in the complete journey for the passengers to get in the bus and for the passengers to drop from the bus .
- Real-time demand thresholds from sensors where the passengers capacity is kept as a threshold.

ADMM handles these constraints in a distributed where each sub problem acts as an individual problem so that those problems is optimized such that it can be applied in a large scale.

Chapter 4

Results and Discussion

This chapter summarizes the key outcomes of the developed intelligent shuttle bus system, covering the LSTM-based passenger forecasting, ADMM optimization for rerouting, hardware implementation for passenger count detection, and a ReactJS dashboard for monitoring and interaction.

4.1 LSTM Model Training and Evaluation

An LSTM model was trained on synthetically generated time-series data to forecast passenger demand at bus stops. Over 50 epochs, the model showed decreasing training and validation loss, converging around 0.0332 and 0.0336, respectively, indicating good generalization.



Figure 4.1: LSTM Training vs Validation Loss

These predictions serve as input to dynamically adjust routing and scheduling decisions in the system.

4.2 ADMM-Based Optimization

To handle the large-scale routing optimization, the Alternating Direction Method of Multipliers (ADMM) was employed. The method breaks the global problem into smaller convex subproblems, enabling efficient parallel computation. This helped in generating optimal bus routes that adapt to demand fluctuations in real time.

```

[DEBUG] ADMM Input - Routes: {'Route A': [0, 1], 'Route B': [2, 3], 'Route C': [4, 5]}
[DEBUG] ADMM Input - Total Buses: 10
[DEBUG] ADMM Input - Distances: [2.5, 3.0, 1.8, 2.2, 3.5, 1.0]
[DEBUG] ADMM Input - Fuel Efficiency: 5.0
[DEBUG] ADMM Allocation: {'Route A': 1, 'Route B': 7, 'Route C': 2}
[DEBUG] Idle Buses: 0
[OPTIMIZER] Updated fleet status.
[DEBUG] Fleet assignment: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0}
[DEBUG] Passenger counts reset to zero
[DEBUG] Received data: {'counts': [1, 4, 5, 42, 9, 9]}
[DEBUG] Reset signal sent to client
[DEBUG] Updated passenger counts: [0, 1, 4, 5, 42, 9, 9, 0]
127.0.0.1 - - [15/Apr/2025 11:28:40] "POST /update_counts HTTP/1.1" 200 -
[DEBUG] Fleet Status Response: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0, 'passenger_counts': [1, 4, 5, 42, 9, 9]}
127.0.0.1 - - [15/Apr/2025 11:28:40] "GET /fleet_status HTTP/1.1" 200 -
[DEBUG] Received data: {'counts': [0, 0, 0, 0, 0, 0, 0, 0]}
[DEBUG] Updated passenger counts: [0, 0, 0, 0, 0, 0, 0, 0]
127.0.0.1 - - [15/Apr/2025 11:28:45] "POST /update_counts HTTP/1.1" 200 -
[DEBUG] Fleet Status Response: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0, 'passenger_counts': [0, 0, 0, 0, 0, 0, 0, 0]}
127.0.0.1 - - [15/Apr/2025 11:28:46] "GET /fleet_status HTTP/1.1" 200 -

```

Figure 4.2: ADMM Optimization Output

4.3 Hardware Setup for Passenger Detection

The hardware system comprises an Arduino Mega and IR sensors placed at bus entry points to count passengers in real time. This data is forwarded for model input and monitoring.

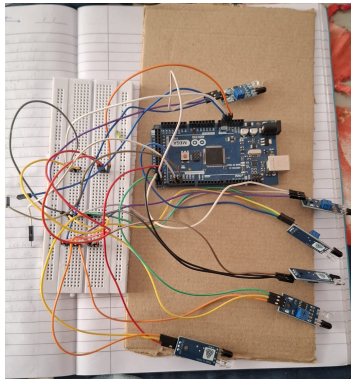


Figure 4.3: Hardware Setup with Arduino and IR Sensors


```
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 3 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
[DEBUG] Raw data from Arduino: DATA:0,1,3,1,1,1
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 3 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
[DEBUG] Raw data from Arduino: DATA:0,1,4,1,1,1
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 4 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
```

Figure 4.4: Passenger Count Display Output

4.4 ReactJS Front-End Dashboard

A web-based interface was developed using ReactJS to visualize live passenger counts, forecast data, and rerouting alerts. It allows system operators to monitor and interact with the model’s output in real time.

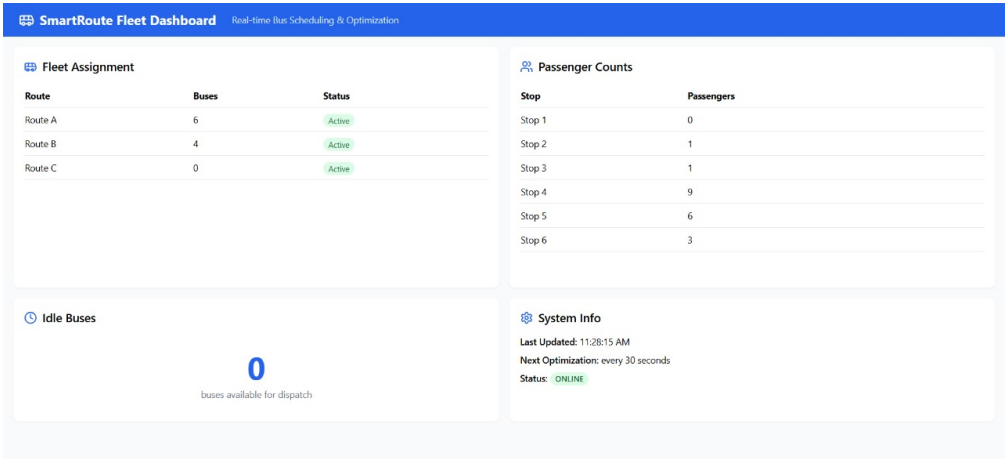


Figure 4.5: ReactJS Monitoring Dashboard

Chapter 5

Conclusion and Future Work

SmartRoute ADMM provides an adaptive solution to urban bus scheduling and resource allocation. By combining AI-driven demand forecasting with ADMM-based optimization, the system enables real-time, demand-responsive scheduling, enhancing passenger experience, service reliability, and operational efficiency. Its scalable network supports dynamic routing, making it ideal for fluctuating urban transit demands.

For future work, we aim to:

- Improve forecasting algorithms using advanced machine learning techniques.
- Explore integration with autonomous vehicles for fully automated operations.
- Incorporate real-time data (weather, traffic, events) for further optimization.
- Expand to include multi-modal transport systems (bikes, metro, etc.).
- Test scalability to handle larger urban areas and more complex routes.
- Optimize energy efficiency with sustainable technologies like electric buses.

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